

Walmart Retail Planning & Scenario Model

Actual vs illustrative plan | January-October 2012 | Selected scenario driven by Assumptions

Selected Scenario **Base**

Model Status **PASS**

Actual Sales
\$1,999.27M

Selected Plan
\$2,008.22M

Plan Variance
(\$8.96M)

Plan Variance %
(0.4%)

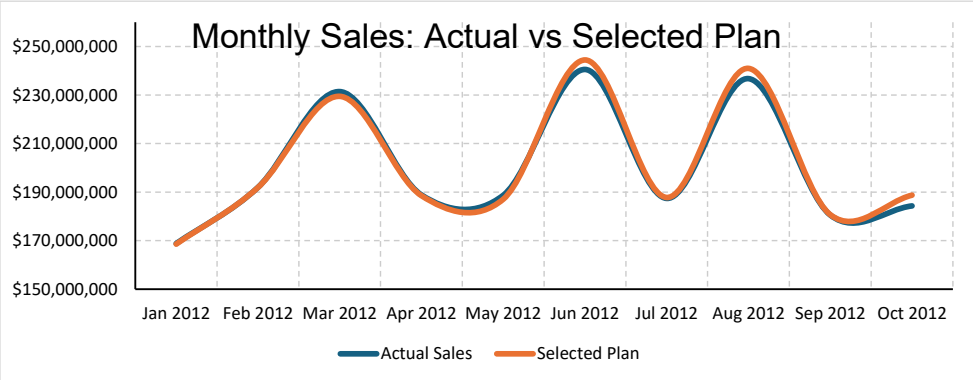
Forecast Sales
\$1,949.73M

Forecast MAE
\$1,812

Favorable Stores
24

Unfavorable Stores
21

Month	Actual Sales	Selected Plan	Variance	Forecast Sales	Variance %
Jan 2012	\$168,804,292	\$168,578,189	\$226,103	\$163,668,145	0.1%
Feb 2012	\$191,945,517	\$191,896,621	\$48,896	\$186,307,399	0.0%
Mar 2012	\$231,415,082	\$229,467,854	\$1,947,228	\$222,784,324	0.8%
Apr 2012	\$188,836,776	\$188,531,139	\$305,637	\$183,039,941	0.2%
May 2012	\$188,685,912	\$187,068,514	\$1,617,398	\$181,619,917	0.9%
Jun 2012	\$240,536,277	\$244,448,951	(\$3,912,674)	\$237,329,079	(1.6%)
Jul 2012	\$187,415,156	\$187,786,328	(\$371,172)	\$182,316,823	(0.2%)
Aug 2012	\$236,756,297	\$240,982,798	(\$4,226,501)	\$233,963,881	(1.8%)
Sep 2012	\$180,553,603	\$180,714,920	(\$161,317)	\$175,451,379	(0.1%)
Oct 2012	\$184,316,563	\$188,747,361	(\$4,430,797)	\$183,249,865	(2.3%)



Walmart Planning Model - Assumptions

Blue-font cells are editable. Plans are illustrative and do not represent Walmart internal budgets.

Scenario	Growth Rate	Holiday Uplift	Purpose
Base	3.0%	0.0%	Illustrative prior-year comparable-week benchmark
Upside	5.0%	1.0%	Stronger growth and holiday performance
Downside	1.0%	-1.0%	Slower growth and weaker holiday performance
Selected Scenario	Base		

Model convention	Value	Units	Year / As Of	Source	Notes
Evaluation period	Jan 6-Oct 26, 2012	Weekly	2012-10-26	Walmart competition data	Matches Power BI report
Planning basis	Prior-year comparable week	Method	2012	Project methodology	52-week lag
Base growth	3.0%	Rate	2012	Project assumption	Editable in B6
Forecast method	Seasonal naive	Method	2012	Python pipeline	Selected after chronological testing
Currency	USD	Dollars	2012	Source data	No inflation adjustment

2012 Scenario Summary

January through October 2012 | Formula-driven from store weekly source

Month	Actual Sales	Forecast Sales	Base Plan	Base Variance	Upside Plan	Upside Variance	Downside Plan	Downside Variance	Selected Plan
Jan 2012	\$168,804,292	\$163,668,145	\$168,578,189	\$226,103	\$171,851,552	(\$3,047,260)	\$165,304,826	\$3,499,465	\$168,578,189
Feb 2012	\$191,945,517	\$186,307,399	\$191,896,621	\$48,896	\$196,096,058	(\$4,150,541)	\$187,697,185	\$4,248,332	\$191,896,621
Mar 2012	\$231,415,082	\$222,784,324	\$229,467,854	\$1,947,228	\$233,923,541	(\$2,508,459)	\$225,012,168	\$6,402,914	\$229,467,854
Apr 2012	\$188,836,776	\$183,039,941	\$188,531,139	\$305,637	\$192,191,938	(\$3,355,161)	\$184,870,340	\$3,966,436	\$188,531,139
May 2012	\$188,685,912	\$181,619,917	\$187,068,514	\$1,617,398	\$190,700,912	(\$2,015,000)	\$183,436,116	\$5,249,797	\$187,068,514
Jun 2012	\$240,536,277	\$237,329,079	\$244,448,951	(\$3,912,674)	\$249,195,533	(\$8,659,256)	\$239,702,370	\$833,907	\$244,448,951
Jul 2012	\$187,415,156	\$182,316,823	\$187,786,328	(\$371,172)	\$191,432,664	(\$4,017,508)	\$184,139,991	\$3,275,165	\$187,786,328
Aug 2012	\$236,756,297	\$233,963,881	\$240,982,798	(\$4,226,501)	\$245,662,075	(\$8,905,778)	\$236,303,520	\$452,777	\$240,982,798
Sep 2012	\$180,553,603	\$175,451,379	\$180,714,920	(\$161,317)	\$184,691,546	(\$4,137,943)	\$176,738,295	\$3,815,309	\$180,714,920
Oct 2012	\$184,316,563	\$183,249,865	\$188,747,361	(\$4,430,797)	\$192,412,358	(\$8,095,795)	\$185,082,363	(\$765,800)	\$188,747,361
Total	\$1,999,265,476	\$1,949,730,752	\$2,008,222,675	(\$8,957,199)	\$2,048,158,177	(\$48,892,701)	\$1,968,287,173	\$30,978,303	\$2,008,222,675

Model Checks and Sources

Check	Actual	Expected	Difference	Tolerance	Status	Where to Fix
Store actuals tie to published total	\$1,999,265,475.88	\$1,999,265,475.88	\$0.00	\$1.00	PASS	Store Weekly
Department actuals tie to store actuals	\$1,999,265,475.88	\$1,999,265,475.88	\$0.00	\$0.01	PASS	Department Weekly
Base plan ties to published total	\$2,008,222,674.93	\$2,008,222,674.93	\$0.00	\$1.00	PASS	Assumptions / Store Weekly
Monthly actuals tie to store actuals	\$1,999,265,475.88	\$1,999,265,475.88	\$0.00	\$0.01	PASS	Scenario Summary
Scenario selector is valid	1.00	1.00	0.00	0.01	PASS	Assumptions B10
No blank scenario assumption cells	6.00	6.00	0.00	0.01	PASS	Assumptions B6:C8

MODEL STATUS

PASS

Item	Value	Units	Period / As Of	Source Type	Source / URL	Notes
Weekly sales source	421,570	Rows	2010-02-05 to 2012-10-26	Competition dataset	https://www.kaggle.com/c/walmart-recruiting-store-sales-forecasting	Raw train, features, and stores files
Evaluation actual sales	\$1,999,265,475.88	USD	Jan-Oct 2012	Python pipeline	python/pipeline.py	Published Power BI total
Base planning scenario	\$2,008,222,674.93	USD	Jan-Oct 2012	Python pipeline	python/pipeline.py	Prior-year comparable week plus 3%
Forecast MAE	\$1,811.56	USD per row	Jan-Oct 2012	Python pipeline	data/processed/forecast_metrics.csv	Seasonal naive
Forecast WMAE	\$1,818.03	USD per row	Jan-Oct 2012	Python pipeline	data/processed/forecast_metrics.csv	Holiday errors weighted 5x
Planning disclaimer	Illustrative	Method	2012	Project methodology	README.md	Not Walmart internal budget or target